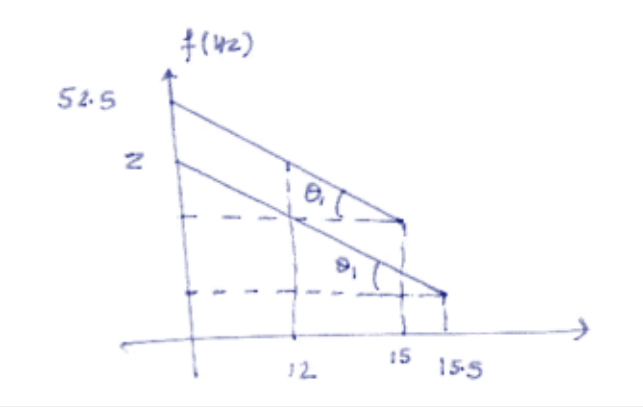
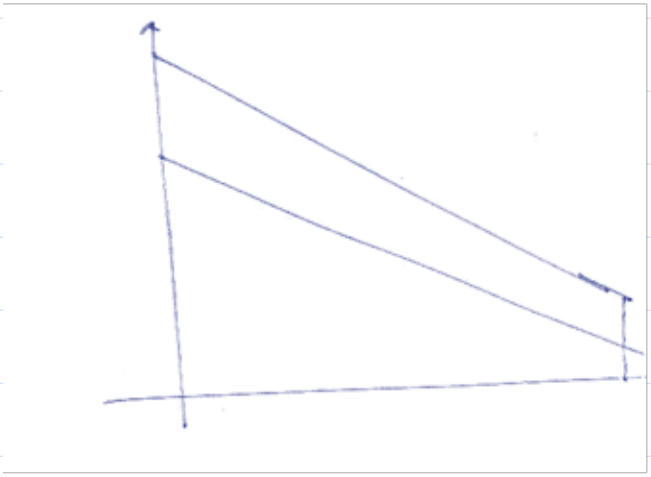
Economic Load Dispatch

DAY # 9 [17/9/25] :

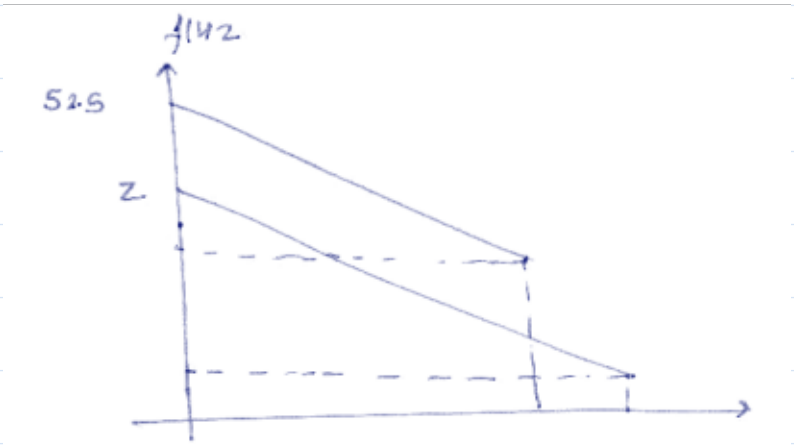


Two parallel droop lines: f (Hz) from 52.5 and 52 against P, with 12, 15 and 15.5 marked, angles theta1

$$K = \frac{2.5}{15} = 0.166$$

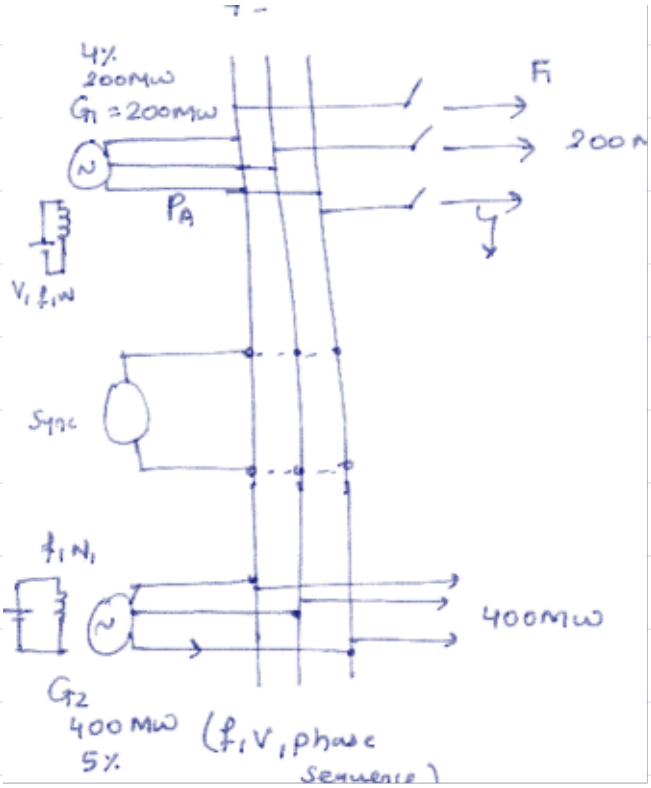


Two droop lines of the same slope shifted by speed changer setting

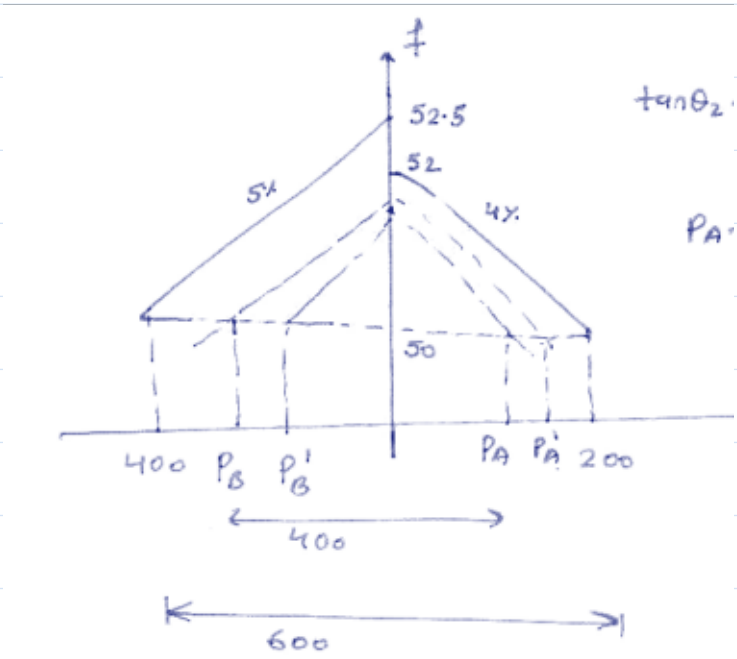


Two droop lines (52.5 and 52 Hz intercepts) with operating points marked by dashed lines

Load sharing of parallel Generators :



Generator G1 = 200 MW, 4 % droop, feeding feeder F1 (200 MW) through the bus; synchronising machine; Generator G2 = 400 MW, 5 % droop (f, V, phase sequence matched) feeding 400 MW



f against P: G2 line 5 % from 52.5 Hz, G1 line 4 % from 52 Hz, common operating frequency 50 Hz, loads PB, PB' and PA, PA', capacity 400 and 200, total 600

$$\tan\theta_1 = \frac{52 - f_n}{P_A} = \frac{52 - 50}{200} = \frac{2}{200} \quad -(1)$$

$$\tan\theta_2 = \frac{52.5 - f_n}{P_B} = \frac{52.5 - 50}{400}$$

$P_A + P_B = 400 \Rightarrow P_B = 400 - P_A$

$$\frac{52.5 - f_n}{400 - P_A} = \frac{2.5}{400}$$

$f_n = 50.77\ Hz\ ;\ P_A = 123\ MW\ ;\ P_B = 277\ MW$

Before tripping of feeder-1 :

$$\frac{P_1}{P_2} = \frac{200}{400} = \frac{1}{2} = 0.5$$

After tripping of F1 :

$$\frac{P_A}{P_B} = \frac{123}{277} = 0.44$$

(2) For economic load Dispatch i.e. to minimize overall cost of Gen, load should be shared in proportion to the Gen capacity.

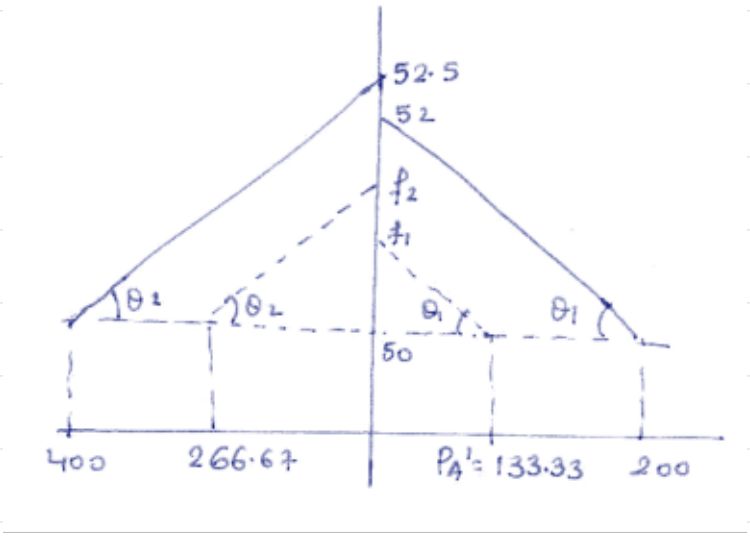
$$\frac{P'_A}{P'_B} = \frac{P_1}{P_2} = 0.5$$

$P'_A + P'_B = 400\ ;\ P'_B + 0.5P'_B = 400 \rightarrow P'_B = 266.67\ MW\ ;\ P'_A = 133.33\ MW$

No load frequencies :

$$\tan\theta_1 = \frac{f_1 - 50}{133.33} = \frac{52 - 50}{200} \rightarrow f_1 = 51.33\ Hz$$

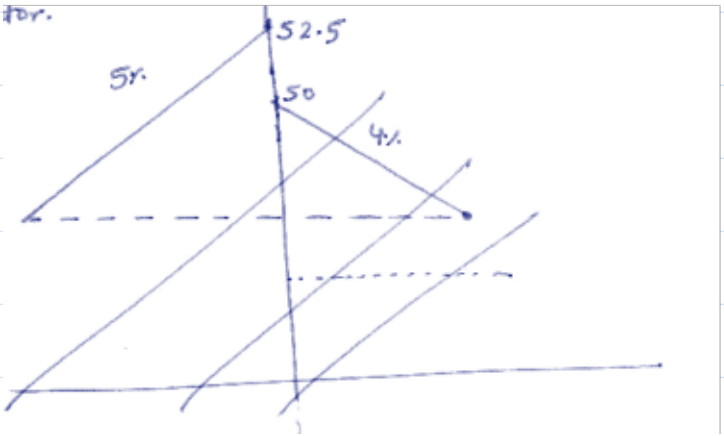
$$\tan\theta_2 = \frac{f_2 - 50}{266.67} = \frac{52.5 - 50}{400} \rightarrow f_2 = 51.66\ Hz$$



Droop lines with no-load frequencies f1, f2 shifted so loads share 266.67 and 133.33 MW at 50 Hz

Eg: Two Generators G_1 and G_2 each having a capacity of 200 MVA operating at no load freq 50 Hz; G_1 has 4 % drooping, G_2 has 5 % drooping ch. For a load of 300 MW find load sharing of G_1 and G_2 and the new operating freq.

(ii) What is the maximum load that can be shared by both the Generators without overloading any of the Generators.



Droop lines for G1 (4 %) and G2 (5 %) from 50 Hz; shifted lines illustrating load sharing

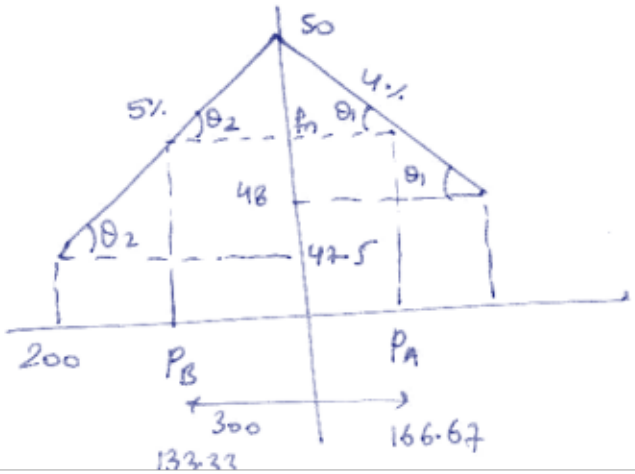
$$\tan\theta = \frac{52 - f_n}{P_A}$$

$\tan\theta_1 = \frac{50 - f_n}{P_A} = \frac{50 - 48}{200}$ -①

$\tan\theta_2 = \frac{50 - 47.5}{200} = \frac{50 - f_n}{P_B} = \frac{50 - f_n}{300 - P_A}$

$P_A = 166.67$ MW

$f_n = 48.34$ Hz



f against P with 50 Hz no-load for both machines: 5 % line and 4 % line meet at common frequency 48 Hz; G2 at 42.5 Hz at 200 MW; PB and PA with total 300

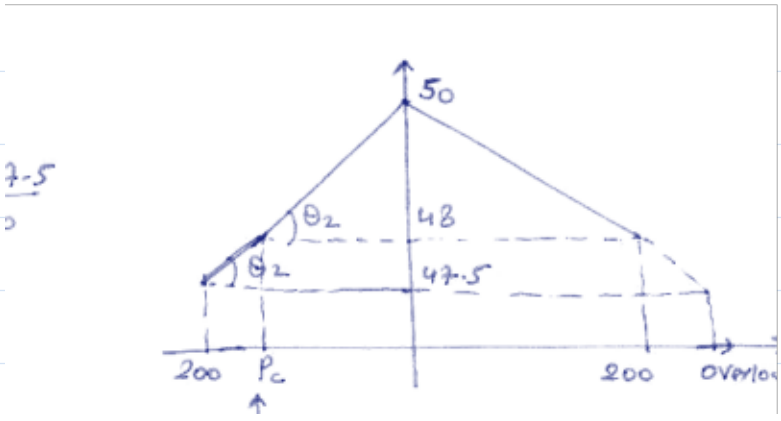
$$\tan\theta_1 = \frac{50 - f_n}{P_A} = \frac{50 - 48}{200} \quad -(1)$$

$$\tan\theta_2 = \frac{50 - 42.5}{200} = \frac{50 - f_n}{P_B} = \frac{50 - f_n}{300 - P_A}$$

$P_A = 166.67$ MW ; $f_n = 48.34$ Hz ; $P_B = 133.33$ MW

$$\tan\theta_2 = \frac{50 - 48}{P_C} = \frac{50 - 47.5}{200} \Rightarrow P_C = 160 \text{ MW}$$

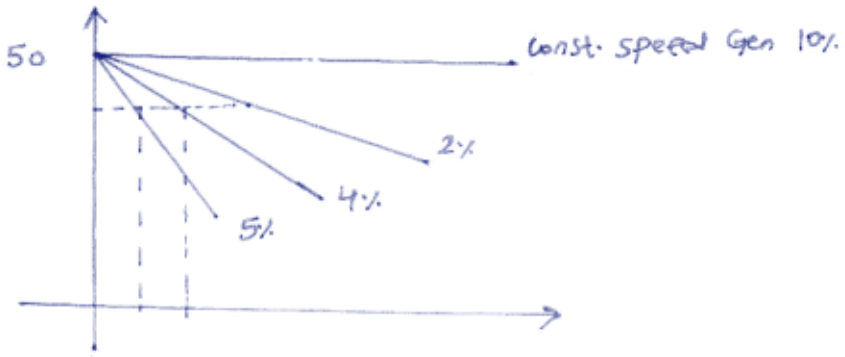
$P_{max} = 200 + 160 = 360$ MW @ 48 Hz (G1 at full 200 MW, G2 at 160 MW)



Droop lines: G2 at 48 Hz carries PC = 160 MW (under load), G1 at 200 MW; overload region beyond; Pmax

* In case of parallel operation of Generators, the Gen which has lowest Drooping ch. will share maximum load.

* If const. speed or 0 % Drooping ch. Generator is connected with the other Generators having 2, 4, 6 % etc. drooping ch., then 0 % drooping ch. generator will share entire load change without loading other parallel Generators, assuming it has infinite capacity.



f against P from 50 Hz: const. speed generator (0 %) is a flat line; drooping lines 2 %, 4 %, 5 %